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The power of big data analytics over fake news: A scientometric review of Twitter as a predictive system in healthcare

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ABSTRACT

Interest in healthcare has grown significantly worldwide, especially since the Covid-19 outbreak. Digitalisation has allowed users to interact on social networks through platforms like Twitter, collecting user interactions over time, resulting in the proliferation of fake news. This research aims to analyse, evaluate and classify the predictive potential of Twitter analytics in healthcare, identifying the latent knowledge insights and distinguishing them from related rumours and fake news. Thus, a systematic literature review (SLR) is carried out to identify and analyse the existing academic research and applications in Twitter in predicting healthcare. The most important predictive applications are detecting mental health issues and public health emergencies. Covid-19 has been the main topic of most of the studies linked to fake news and misinformation, and this research provides a practical contribution to the use of unstructured data from Twitter and raises awareness of the importance of this content applied to healthcare. Therefore, it is pertinent to focus on the advances offered by these data as a predictive tool in healthcare since it is essential, to this end, to evaluate the veracity of the information shared on Twitter.

1. Introduction

The use of social networks has grown dramatically in recent years. The growth in the use of social platforms globally has greatly increased the amount of data available to monitor and predict trends in society, economy and business, and healthcare (Burnap et al., 2016). Twitter is a relevant source for obtaining up-to-date information from individuals and organisations, which enables the development of heterogeneous studies in multiple subjects. These studies, relying on the analysis of the aggregated data and the social networks generated among users, allow us to understand the social perspective of the given research topic, based on users' perceptions and feedback. Social media platforms are characterised by the rapid spread of misinformation, often known as fake news (Wang et al., 2019).

The world is awash with data being generated increasingly quickly as information technology expands in society (Alazzawe et al., 2020). The sheer volume, easy access and rapid spread of online information make social networks a perfect example of social interaction between millions of individuals (Boyd and Crawford, 2012). In line with Wamba et al. (2015), the analysis of this vast amount of information is generally called big data analytics, defined as a holistic process for managing, processing and analysing data, characterised by its volume, variety, velocity, veracity and value (the '5 V's'). Big data analytics aims to create business value, measure performance, and generate competitive advantages by improving decision-making (Wamba et al., 2017).

The information revolution has given rise to new economies structured around data, information and knowledge flows (Serrat, 2017). Information consumption and access to information are changing and evolving as society does, and social media is considered a key channel in digital, as it allows greater access to information through transparency and accountability (Andersen et al., 2012). Microblogging has become one of the most useful tools for sharing events and news from everyday life and expressing opinions about those events (Ridhwan and Hargreaves, 2021). With the development of social media platforms, more and more people post online texts on different platforms to express their opinions on social issues (Zhang et al., 2012), thus turning the user into a content generator and leading to the construction of networks around the same topic. Users of online social media share information, interact with other users and stay informed, but they also generate dubious information, sometimes intending to mislead (Zhang and Ghorbani, 2020). This information is considered "fake news", as it refers to social media posts with fabricated, misleading or distorted information deliberately intending to misinform (Casillo et al., 2021). The internet has changed how people communicate, significantly influencing how people and companies interact (Tembhurne et al., 2022). In recent years, the rise of online social networks has led to the

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proliferation of social news. Many social platforms, such as Facebook, Instagram and Twitter, are affected by the sharing of fake news among users (Sahoo and Gupta, 2021).

Web 2.0 tools, such as Twitter and various blogs, allow us to collect new metrics from other sources that provide data in a structured format, accessible through application programming interfaces (APIs) (Priem and Hemminger, 2010). Among microblogging platforms, Rathore et al. (2017) showed that Twitter is the most popular social media platform due to its broad applications. Social network analysis (SNA) is a well-established method that focuses on relational patterns among "actors" (e.g., individuals, institutions, nations) (Li et al., 2022). As Scott (1988) proposed, the central question in the development of social network analysis has been how the metaphor of a social network should be taken. It describes the networks and connections between individual or collective actors, embedded in webs of connections, and the main task is to describe and explain the derivatives of these connections. The authors have categorised the healthcare field inductively, using groups based on the basic concepts and definitions proposed by Stieglitz et al. (2018).

In recent years there have been studies related to the use of Twitter from different perspectives, considering topics such as ethnicity/regional origin (Pennacchiotti and Popescu, 2011), political orientation (Gayo-Avello, 2012), digital marketing and business strategies (Piñeiro-Chousa et al., 2021), and authors such as Sie et al. (2012) have conducted literature reviews that consider social network analysis (SNA) techniques from a global perspective. But its influence on healthcare has not yet been widely considered. In the case of fake news, Tandoc (2019) performed a systematic review on how to categorise and agree on how to define fake news and misinformation, how these are generated, how people are prone to their impact and how to minimise this influence and its effectiveness. In the area of health, Rocha et al. (2021) elaborated a detailed review about the spread of misinformation during the Covid-19 pandemic, showing that psychological disorders, panic, fear, depression, and fatigue were generated in the people exposed to this unreliable content. These findings agreed with those of Alowibdi et al. (2021), who performed a topic and sentiment analysis of the social perception of this coronavirus pandemic on Twitter. The present authors have not found any studies offering in-depth and holistic analysis of Twitter analytics and fake news in the health field.

Therefore, this study aims to fill this gap in the literature about the predictive use of Twitter analytics applied to healthcare and the role of fake news. The SLR methodology also allows emerging issues to be considered (Bem, 1995). The analysis of existing literature in this area is important because it allows the generation of new research and the ordering of concepts generated up to the present time according to their representativeness. Additionally, and following Gaurav et al. (2021), who emphasise that although people get information online, some of the news and comments on the internet are questionable, it is therefore valuable to analyse the impact of two types of scientific articles: those that propose Twitter as a predictive system in healthcare and those that highlight the influence of this platform as a way to share information of dubious validity. The aim is to analyse the functionalities and benefits identified by the authors over the years and consider the future lines of research that are derived from them.

In light of the above, the following research questions are posed:

RQ1: How important is Twitter as a predictive tool in the area of healthcare?

RQ2: With which topics has Twitter been linked in the healthcare field?

RQ3: How has interest in the use of Twitter as a forecasting system in global healthcare evolved?

RQ4: What is the impact of fake news in predicting healthcare through Twitter analytics?

To address these research questions, the rest of this article is organised as follows. Section 2 presents a theoretical framework that positions the relevance of Twitter as a predictive system in the health area, and Section 3 explains the methodology (Section 3) used. The results are presented in Section 4 and discussed in Section 5. Finally, the main contributions of the present research are described in the Conclusions section, including the theoretical and practical implications, the evaluation of the limitations of the study and the presentation of potential new future lines of research.

2. Theoretical framework

In line with the World Health Organization (WHO)'s proposal, one of the main functions of public health work is the evaluation and monitoring of the health of communities and populations, particularly those at risk, in order to identify health problems and priorities. Various authors have proposed Twitter as a tool capable of providing a real and updated view of the content related to this topic, generating great advances in the area of health. It has been used to predict and track diseases such as Ebola (Van Lent et al., 2017), H1N1 flu (Chew and Eysenbach, 2010), Covid-19 (Hammer et al., 2021), mental health issues (Bouarara, 2021), including depression and suicidality in the population (O'dea et al., 2015) or the dynamics of seasonal and non-seasonal influenza outbreaks, e.g., one week in advance (Lipsitch et al., 2011). This type of study became especially important with Covid-19, which has affected the worldwide population, generating a social impact through social distancing and quarantines, beyond healthcare (Chopra et al., 2022). Social networks have acquired special prominence since they can be used both to measure the current state of a specific disease and to predict possible infections and thus help to alert the population and share information in real-time. Harmful behaviours can be identified from user reports of negative consequences that may require professional intervention (Kuss et al., 2013). It has also been proposed by scholars such as Samaras et al. (2020) that ICT platforms like Twitter can influence the evolution of smart healthcare systems. For example, Twitter has created internal mechanisms that see when individuals express content with suicidal overtones and has enabled channels to report it O'dea et al. (2015). This is a breakthrough in the use of the tool as a predictive system in public health and mental health.

Most of the unstructured data are generated through social platforms (Gandomi and Haider, 2015), and Twitter offers a large volume of unstructured data generated from heterogeneous sources in a short space of time (Udanor et al., 2016). It provides the user with a platform on which to interact and generate networks of contacts freely, since it is a free broadcast social network that allows registered users to communicate with others in real time using 280 characters, originally 140 characters. The tweet is a key piece of information: it contains the user name and identifies the publication by its status and timestamp (Achrekar et al., 2011). This, alongside other analytics, allows the data to be used scientifically. Through the social network itself, the user's concerns can be observed based on the content they generate. Twitter content can be published through a web interface, an SMS or a mobile device, which is why the immediacy and dynamism among users to create and share content has continuously grown since its creation. The number of social network users is increasing, particularly teenagers accessing mobile internet technology through smartphones, laptops and tablets (Lenhart et al., 2010). There has been a corresponding remarkable increase in the amount of accessible data, from which important conclusions can be drawn that influence the development of society. In order to collect these data, the platform must be monitored using Twitter's Application Program Interface (API).

Bearing in mind the analysis of information linked to social networks, it should be noted that there are two main analyses. The first is SNA, which looks at how users interact and connect. The second is natural language processing (NLP), which allows the textual content of tweets to be analysed (Cossu et al., 2016) automatically. Between them, these two approaches have offered many possibilities for research,

and many scientific works have used them to draw conclusions from analysing big data from Twitter.

Twitter as a predictive system can contribute to infodemiology (a portmanteau of information and epidemiology), a science that considers the distribution and determinants of information in the electronic medium (Eysenbach, 2009). Twitter enables the monitoring of updates on the status of individuals and can thus be useful in identifying and monitoring relevant health-related publications.

Traditional methods of collecting information related to healthcare require much more time, making them more costly and less immediate, so some new considerations might be missed. Also tweets, in combination with other data sources such as incidence rates, can provide high-impact information (Chew and Eysenbach, 2010), and Twitter as a predictive system has been used in various scientific works. Special importance has been attached to research work referring to fake news because – despite its being difficult to detect (Sahoo and Gupta, 2021) –, it is especially important in today's world. It is intentionally disseminated to mislead readers and is the major challenge in identifying the quality of information exchanged through social platforms.

The current study analyses how Twitter is used as a predictive tool and how false or malicious information in the tweets can be identified. In such research the information from Twitter needs to be considered alongside secondary data sources and the social context from which it emerges.

Data from Twitter, used in this way, is a highly potent digital tool for detecting both truthful and fake information related to health care, forecasting the future of healthcare and creating value with truthful content.

3. Methodology

The SLR was conducted following the SPAR-4-SLR framework (Paul et al., 2021), and the analyses proposed by other authors such as Ribeiro-Navarrete et al. (2021) and Martínez et al. (2022), in which the concepts are structured to generate a coherent review of the topic itself. The technique for the review uses a completely and transparently reproducible end-to-end process consisting of three phases, called 'the 3 As': assembling, arranging and assessing. 'Assembling' is the process of acquiring and consolidating literature from different data sources. 'Arranging' is the processing and organising of the literature, and 'assessing' is the evaluation and reporting of the literature in the scope of the study. The methodology is described in Fig. 1.

There are three main outcomes of this SLR study of the use of Twitter analytics as a predictive tool in healthcare. The first is the understanding and characterisation of the state-of-the-art in this field. The second is the identification of research opportunities and practical applications. The third is guidance on how to overcome fake news in this domain.

3.1. Data collection and sample

Starting with the assembling phase of the SLR, Fig. 1 shows the search procedure used to carry out the research. Therefore, the search terms used were:

$$ST = (\text{“}Twitter\text{”}\ OR\ \text{“}tweet\text{”})AND(\text{“}predict\text{”}\ OR\ \text{“}forecast\text{”}) \quad (1)$$

The query is defined to identify publications about Twitter as a predictive system, not only specifically related to healthcare but in general. This is due to the wide range of its applications in predicting healthcare (detection of adverse events, epidemiology, mental health, etc.), the lack of existing studies related to the review of the use of Twitter as a predictive system in healthcare, and the myriad applications of Twitter data to make predictions in multiple areas of knowledge, such as economy, politics or society. Inserting general terms such as "health" or "healthcare" would have biased the search and limited its scope, since the keywords represent critical information in the target research area. A subsequent manual screening was therefore required to limit the scope to healthcare. In order to maximise scientific solvency, a database was constructed from a wide variety of scientific papers, articles, reviews and conference papers indexed in the well-known academic databases of Scopus, IEEE, Web of Science, ScienceDirect and Dimensions.ai (Singh et al., 2021), covering the highest impact journals and conferences. In total, 7928 publications that considered the predictive potential of the tool were collected, dated between March 2006 (when Twitter was launched publicly) until November 2021.

The exclusion criteria applied in the arranging stage of the SLR are as follows. The duplicates and publications with invalid DOIs were eliminated from the scope of the study, reducing the scope to 4465 papers. After a manual screening conducted consensually by all three authors, who agreed on the classification, the publications focused mainly on the use of Twitter as a predictive tool in the area of healthcare were selected. Those focused on other fields of study were discarded, as also were studies whose purpose was not to consider Twitter as an analytical or predictive system. The three authors have consensually reviewed and agreed on this classification. After this arranging process, 492 publications within the scope of the present research remained as the sample of our study. In a nutshell, the articles leveraging the predictive potential of Twitter analytics applied to healthcare are in line with the defined inclusion criteria.

4. Results

4.1. Content analysis

As part of the assessing phase of the SLR, to address the RQ1, a large-scale content analysis was performed using Gephi, a specialised software for network analysis, and Biblioshiny (Aria and Cuccurullo, 2017), a bibliometric tool based on the Bibliometrix library and R software. A topic identification algorithm based on network analysis (Paranyushkin, 2019) was applied to all the papers in the scope of the research (492 publications). A text-to-network graph is built, where the nodes are the unique lemmas of the corpus formed by the set of titles and abstracts of the publications in scope, and the edges are defined by the co-occurrences of the lemmas. The edge weight varies from three to one, depending on the distance between concurrent lemmas, which varies from one to three lemmas of distance, as defined in the pseudo-code in Algorithm 1.

In this analysis, Gephi, a visualisation and analytical software for graphs and networks, was used for the network analysis of the generated graph. The force-directed graph layout algorithm called Yifan Hu Multilevel (Hu, 2005) was applied to the generated text-to-network graph. The force-directed graph layout algorithms – outlining Fruchterman and Reingold (Fruchterman and Reingold, 1991), Force Atlas (Noack, 2007) or Force Atlas 2 (Jacomy et al., 2014), which was applied originally by Paranyushkin (2019) – model a graph as a physical system of bodies, existing forces attracting and repelling the nodes. The goal of the algorithm is to minimise the energy of the system in its convergence. This Yifan Hu Multilevel layout algorithm is very similar to the Yifan Hu layout algorithm (Hu, 2005), which, as a force-directed algorithm, computes the new layout of the network by optimising the overall internode repulsions. Unlike the Force Atlas algorithm, it only considers the pairs of adjacent nodes in the calculation and thus performs better, while its repulsion forces produce a more understandable visualisation in this case. The Yifan Hu layout algorithm also applies the 'adaptive cooling scheme', allowing faster convergence and preventing the algorithm from getting stuck in local minima. The Yifan Hu Multilevel differs from the Yifan Hu graph layout algorithm in that it uses a proportional displacement strategy for node placement in the graphical space, reducing computational complexity while keeping

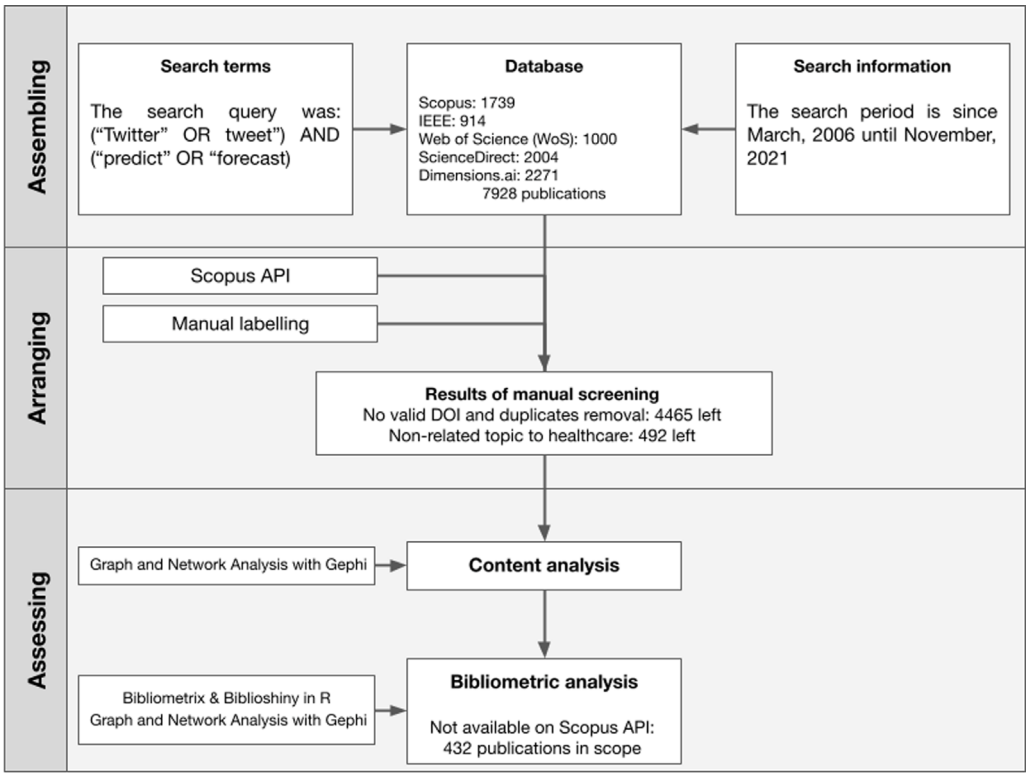

Fig. 1. SPAR-4-Systematic literature review procedure.

Algorithm 1: Algorithm for text-to-network graph generation

Result: $G(V, E)$ is the directed graph G formed by the set V of vertices, which represents lemmas, and the set E of edges that represent collocations between the lemmas in the corpus.

Data: corpus

```
maxdistance ← 3
foreach sentence ∈ corpus do
    sentence ← preprocess(sentence)
    sentence ← lemmatise(sentence)
    foreach position, lemma ∈ enumerate(sentence) do
        distance ← 1
        weight ← maxdistance
        G.add_node(lemma)
        while (distance < (length(sentence) − position) and
         distance ≤ maxdistance) do
            if G.has_edge(lemma, sentence[position + distance])
             then
                G[lemma][sentence[position +
                 distance]]['weight'] += weight
            else
                G.add_edge(lemma, sentence[position +
                 distance], weight)
            end
            distance += 1
            weight −= 1
        end
    end
end
```

similar accuracy and speed. To create the final representation of the graph in Fig. 2, the Label Adjust algorithm (Cherven, 2015) is used to prevent the overlapping of the tags of the nodes and to filter the network by node degree higher than 150 and the edge weight higher than 20.

Finally, to complete the topic identification process (Table 1), the betweenness centrality parameter (C_B) is calculated for each node in the co-occurrence network, as described in Eq. (2) (Freeman, 1977), and then normalised between 0 and 1:

$$C_B = \sum_{s \neq t \neq v \in Vertices} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

Given a directed graph G(V,E), formed by a set V of vertices and a set E of edges, where σ_{st} is the number of shortest paths from s to t, with $s, t \in V$ and $\sigma_{st}(v)$ denotes the number of shortest paths from s to t passing through the node $v \in V$.

Then, the nodes are grouped by modularity class (Blondel et al., 2008) so that the top keywords of each modularity class will allow the inference of each related topic, analogously with the case of various topic identification algorithms such as Latent Dirichlet Allocation (LDA) (Blei et al., 2003).

As a result of the topic identification analysis, mental health was identified as the most representative topic, having an outlined presence with 29.56% of the lemmas belonging to this class. Suicide (Birjali et al., 2017), depression (Ziwei and Chua, 2019), food disorders (Abbar et al., 2015) and addictions (Eshleman et al., 2017) are mental health issues widely studied on Twitter for both detection and prediction, including by authors such as Coppersmith et al. (2014). Given the increasing use of social platforms for creating awareness about mental health (Saha et al., 2019), it is possible to improve the available data in the field of mental health, allowing the massive analysis of a high volume of content in Twitter through natural language processing (NLP) methods.

Healthcare surveillance on social media should be highlighted; authors such as Gupta and Katarya (2020) have shown how social networks are used as surveillance systems, thus being able to collect

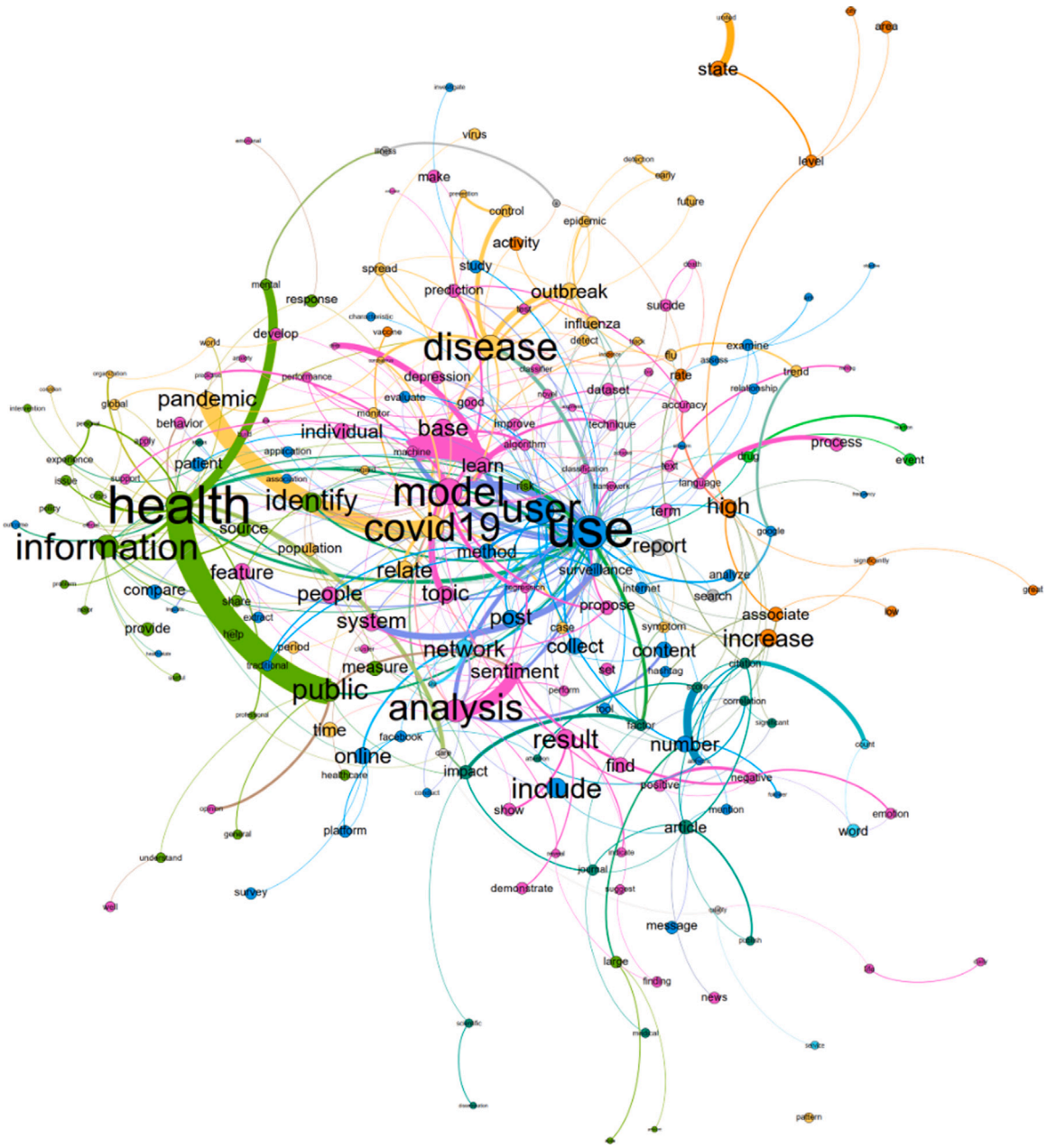

Fig. 2. Co-occurrence network for the corpus of titles and abstracts.

and analyse the available information to predict the evolution of diseases. Jordan et al. (2018) also used Twitter as a tool to monitor and predict public health issues.

The work of St Louis and Zorlu (2012) links topic 4 and the use of Twitter in predicting disease outbreaks such as the seasonal flu, and monitoring and predicting the evolution of epidemics: especially relevant during the Covid-19 pandemic. The platform also became an important source for infodemiology by collecting comments, opinions and emotions from individuals around Covid-19 (Xue et al., 2020).

Next, topic 5 offers the perspective of health in society, as proposed by Ireland et al. (2015) in the case of HIV. Similarly, in the case of Covid-19, Twitter was used for predicting vaccination uptake and social perspectives on it Zhang et al. (2021).

The next theme, Bibliometrics, focuses on the observation and prediction of the relevance of the scientific output and the different forms of impact. One example is Altmetrics, alternative citation metrics based on web 2.0 for evaluating the scientific impact and relevance of academic publications (Priem et al., 2011), using it, as a practical example, to measure the effects produced in clinical practice (Chang et al., 2019).

Taking into account the impact of new technologies, topic 7 reflects the importance of the predictive capabilities of artificial intelligence techniques, highlighting machine learning and deep learning algorithms like neural networks, specifically Long Short-Term Memory (LSTM), applied to healthcare. It is worth noting two particular contributions. Khan et al. (2020) used these machine learning methodologies to predict the sentiment of the user-generated content during Covid-19, and Jayawardhana and Rajapakse (2020) presented an LSTM architecture to predict disease outbreaks through the use of public data from Twitter.

Finally, topic 8 refers to the patient himself/herself and pharmacovigilance or patient safety activities, through the identification of adverse drug reactions, in line with what is proposed by O'Connor et al. (2014).

Table 1
Inferred topics with text-to-network analysis.

		%	Topic	Keywords
1		29.56	Mental health	Model, Depression, Prediction, People, Sentiment, Positive, Negative, Life, Anxiety, Behaviour, Suicide, News, Death, Suicidal, Observe
2		16.96	Public health	Health, Information, Identify, Public, Measure, Source, Community, Provide, Response, Concern, Share, Risk, Large, Help, Communication
3		16.95	Health surveillance	Use, User, Include, Post, Online, Number, Collect, Content, Method, Compare, Patient, Surveillance, Study, Message, Follow
4		13.05	Disease outbreak. Focus on Covid-19	Covid-19, Disease, Pandemic, Relate, Outbreak, Time, Influenza, Population, Period, Control, Virus, Spread, Case, Trend, Flu
5		7.26	Social perspective on health	Increase, High, State, Associate, Activity, Rate, Level, Area, Low, Food, Great, Vaccine, Group, City, County
6		6.02	Bibliometrics	Article, Impact, Factor, Citation, Journal, Correlation, Score, Significant, Medical, Discussion, Range, Publish, Scientific, Publication, Influence, Sequence, Memory
7		2.33	Artificial intelligence techniques in healthcare	Network, Word, Count, Service, Site, Alone, Program, Artificial, Neural, Embed, Urology, Long, Few, Sequence, Memory
8		1.53	Patient safety	Effect, Event, Drug, Reaction, Name, Place, Adverse, Interaction, Medication, Addiction, Market, Abuse, ADR, Therapy, Entity

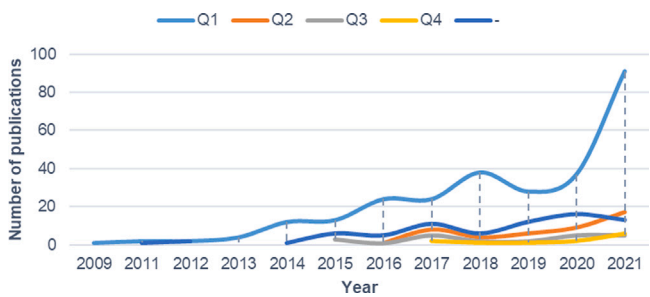

Fig. 3. Evolution of the number of publications per journal impact factor.

4.2. Bibliometric analysis

The last step of the present SLR, as part of the assessing phase in the SPAR-4-SLR framework, was the bibliometric analysis. This identified the most cited papers and authors, along with the main countries and affiliations in scientific production and the most relevant journals. To this end, the Scopus API was used to extract the citation data (Rose and Kitchin, 2019), in order to have a consistent and harmonised data source (Montoya et al., 2018). 432 papers remain in the scope of the study (87.8%, out of 492 papers): sixty publications not available in this bibliometric database were discarded.

To answer RQ2, Fig. 3 shows the evolution in scientific interest in the use of Twitter as a predictive system in the healthcare area from 2009 until 2021, using the Bibliometrix time cut-off function (Aparicio et al., 2019) combined with the JCI impact factor index (SCImago, 2022). Twitter as a predictive system in healthcare is shown to have acquired relevance and increasing representation in Q1 journals. In line with what is proposed, the papers indexed in Q1 journals are also the most considered in other scientific works. The 91% growth observed in 2021 (from 69 articles in 2020 to 132 in 2021) stands out. It is also worth mentioning that 69.86% (51 out of 73 publications) of publications without impact factor (-) are conference proceedings, which reflects the fact that, after an increasing trend, there was a stagnant growth in 2020 and 2021, mainly due to the large numbers of conferences being cancelled and virtualised during the Covid-19 pandemic (Moss et al., 2021).

At a general level, aligned with the aforementioned identified topics through the text-to-network analysis (Table 1), the trending topics reflected on the set of abstracts are represented in Fig. 4. Thus, the initial studies on using Twitter to predict public health events, such as disease outbreaks and their evolution, are identifiable with keywords like “assessment”, “influenza”, “flu”, “ili” (influenza-like illness), “people” or “surveillance”, among others. Mental health is represented with keywords such as “suicidal” or “population”. Finally, the appearance of the “covid” and “pandemic” topics, starting in 2020, assumes special relevance as the Covid-19 outbreak was the most severe public health emergency in recent years, with a big impact on society and the economy (Ahmad et al., 2020), as reflected in the academic literature.

As the healthcare area continues to evolve in facing the challenges presented by globalisation, the investigation of the flow of topics can be considered. For the evolution of topics, the Sankey Plot in Fig. 5 was designed with Bibliometrix. In this case, the length of the block refers to the frequency of the bi-grams in the abstracts of the publications in the scope of the study. This is an important aid for pointing out the linked topics in the use of Twitter as a predictive system in healthcare. Thus, the combination of public data from Twitter, public health and flu trends are key features with concrete targets in the use of Twitter as a predictive system that evolves over the years to mental and personal health and deep learning, highlighting the technology evolution in recent years. The interrelation of these key themes enables a framework that tracks the use of Twitter as a predictive system related to health, thus providing alignment of topics and increasing interest in it and identifying those that have lost interest over time.

With regard to the most relevant scientific academic research, Table 2 includes the ten most cited papers since 2006. It is worth highlighting the most cited article by Signorini et al. (2011), with 768 citations, which refers to the H1N1 influenza A pandemic using Twitter as an effective tool in the U.S. to monitor the evolution of the swine flu and measuring the social perspective and the sentiment with respect to the disease itself. This is followed by the articles proposed by Chew and Eysenbach (2010) and Eysenbach (2009). The former explores the feasibility of measuring the social impact and public attention of academic articles by analysing social networks with 658 citations in 2011, and the latter, with 610 citations, considers Twitter as an infodemiology and infoveillance tool. It is important to emphasise that this article (Chew and Eysenbach, 2010) was a pioneering work in the area of research, setting the path for future research, as it was the first publication to refer to Twitter as a predictive system, not only in the area of health but in any area of study. The article by Achrekar et al. (2011) has 326 citations due to the relevance of its discussion of using Twitter data for making predictions about influenza outbreaks.

Apart from the most cited publications, described in Table 2, Table 3 also shows the papers with the highest citation potential in the coming years, based on the current number of times cited and their recent publication since 2020. Thus, Simcock et al. (2020) summarise the discussions held on Twitter around Covid-19 infection prevention of oncology patients. In addition, the previously-mentioned presence of

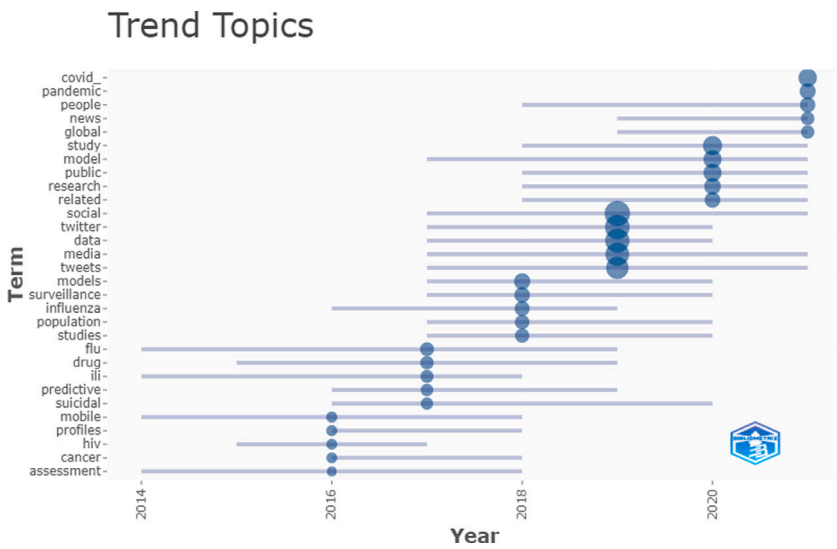

Fig. 4. Evolution of trend topics in abstracts.

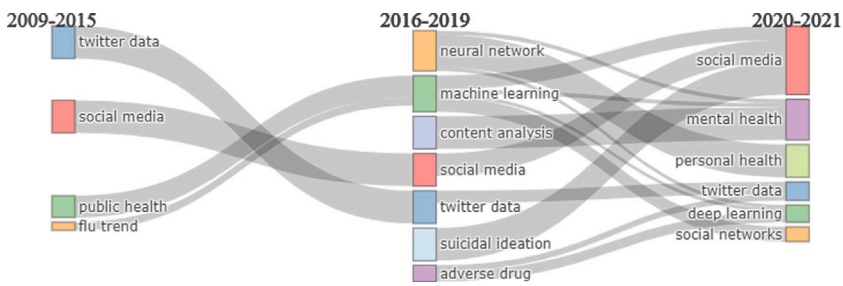

Fig. 5. Thematic evolution with Bibliometrix.

Table 2
Top cited publications.

Title	Authors	Year	Times cited
The use of Twitter to track levels of disease activity and public concern in the U.S. during the influenza A H1N1 pandemic	Signorini, A., Segre, A. M., & Polgreen, P. M	2011	768
Can tweets predict citations? Metrics of social impact based on Twitter and correlation with traditional metrics of scientific impact.	Eysenbach, G.	2011	658
Infodemiology and infoveillance: Framework for an emerging set of public health informatics methods to analyze search, communication and publication behavior on the internet	Eysenbach, G.	2009	610
Predicting flu trends using Twitter data	Achrekar, H., Gandhe, A., Lazarus, R., Yu, S. H., & Liu, B.	2011	326
Internet addiction in adolescents: Prevalence and risk factors	Kuss, D. J., Griffiths, M. D., & Binder, J. F.	2013	252
Combining search, Social media, and Traditional data sources to improve influenza surveillance	Santillana, M., Nguyen, A. T., Dredze, M., Paul, M. J., Nsoesie, E. O., & Brownstein, J. S.	2015	230
Detecting suicidality on Twitter	O'dea, B., Wan, S., Batterham, P. J., Calear, A. L., Paris, C., & Christensen, H.	2015	210
Detecting depression and mental illness on social media: an integrative review	Guntuku, S. C., Yaden, D. B., Kern, M. L., Ungar, L. H., & Eichstaedt, J. C.	2017	196
Discovering health topics in social media using topic models	Paul, M. J., & Dredze, M.	2014	159
You tweet what you eat: Studying food consumption through Twitter	Abbar, S., Mejova, Y., & Weber, I.	2015	145

Table 3
Potentially impactful publications.

Title	Authors	Year	Times cited
COVID-19: Global radiation oncology's targeted response for pandemic preparedness	Simcock, R., Thomas, T. V., Estes, C., Filippi, A. R., Katz, M. S., & Pereira, I. J.	2020	131
Does tweeting improve citations? One-Year Results From the TSSMN Prospective randomized trial	Luc, J. G., Archer, M. A., Arora, R. C., Bender, E. M., Blitz, A., Cooke, D. T., ... & Antonoff, M. B.	2021	64
Who do you trust? The digital destruction of shared situational awareness and the COVID-19 infodemic	Bunker, D.	2020	49
An exploratory study of COVID-19 misinformation on Twitter	Shahi, G. K., Dirkson, A., & Majchrzak, T. A.	2021	36
COVIDSenti: A Large-scale benchmark Twitter data set for COVID-19 sentiment analysis	Naseem, U., Razzak, I., Khushi, M., Eklund, P. W., & Kim, J.	2021	21

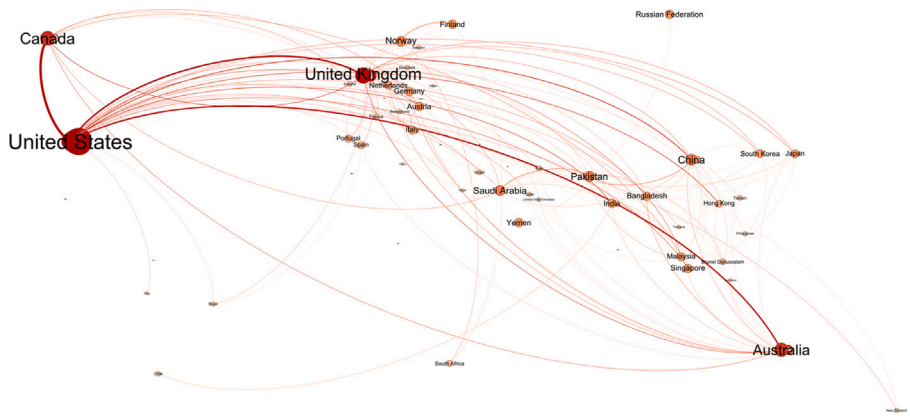

Fig. 6. Collaboration network between affiliations by country.

the Covid-19 topic in the top 5 potentially impactful publications can be seen again, except for the bibliometric study of Twitter in predicting citations by Luc et al. (2021).

4.3. Affiliation collaboration and journal analysis

The collaboration network between countries shown in Fig. 6 was processed in Gephi. The nodes are the home countries of the affiliations of the authors of the publications, with size defined by PageRank (Page et al., 1999). This is an algorithm developed by Google used to rate Web pages and applied in this case to identify the impact of the authors and affiliations (Yan and Ding, 2011). The graph is represented by a geo-layout, having extracted the coordinates (latitude and longitude) of the location of each country. In addition, a filter for the edges whose weight is lower than five is applied, so that those collaborations between affiliations with five or more partnerships are represented. The network has 56 nodes, which refer to affiliations from 56 different countries and 161 links between them, representing the collaborations in place between affiliations. In this case, the importance of the United States, the United Kingdom, Canada and Australia stands out. The United States has the highest number of contributions that link Twitter as a predictive system in healthcare (1051), 27 different collaborations with abroad affiliations, especially from Canada, and the greatest impact with a PageRank of 0.167. Second is the UK, with a PageRank of 0.096, 255 publications and collaborations with institutions from 23 different countries. Lastly, the contributions from China affiliations and authors are worth mentioning, given that Twitter has been officially blocked in China since 2009 as part of government censorship after the social unrest in the Xinjiang area (Xiguang and Jing, 2010).

Analogously, in Fig. 7, the affiliation collaboration network is represented with Gephi using the geo-layout, representing the network geographically, with the nodes being the coordinates of the affiliation they represent. The size of the nodes is determined by the aforementioned PageRank value; a degree filter excluded affiliations whose degree is lower than 13, so only those affiliations with at least 13 contributions are shown. The network consists of 742 different nodes or affiliations, with 1178 edges or collaborations between affiliations. In this case, Harvard Medical School, with 19 contributions and a PageRank of 0.162, is the most impactful university in using Twitter to predict healthcare. Close behind is Children's Hospital Boston (18 unique publications and 0.154 PageRank) and third, further behind, is the University of Melbourne (4 unique publications and 0.09 PageRank).

Table 4
Scientific production by country and affiliation.

Country	NP	TC	Affiliations	NP
US	795	4530	University of Pennsylvania	31
UK	138	941	Harvard Medical School	25
Australia	119	737	University of Milan	22
India	97	609	Arizona State University	20
China	85	217	Deakin University	19

Table 4 summarises the top 5 countries, affiliations and authors in terms of academic relevance in the present research. In this case, it should be noted that the U.S. is the country of maximum scientific relevance, with the University of Pennsylvania, Harvard Medical School and the University of Milan as the most important affiliations according to the total number of publications (in this case, the criterion for counting the number of works is different, deeming each author from the affiliation involved in each paper and not unique publications), using the affiliation and author analysis functionalities in Bibliometrix.

Table 5 shows the result of the analysis of the journals and conferences with the highest scientific relevance and production. The Journal of Medical Internet Research, PLoS ONE, Computers in Human Behavior and the Journal of Biomedical Informatics are the most important journals, accumulating 9.75% of the total publications, while Procedia Computer Science is the most important conference series.

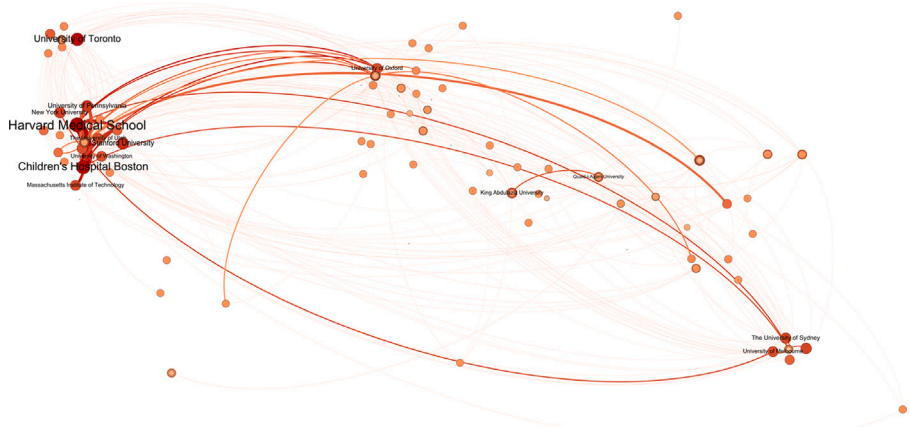

Fig. 7. Collaboration network between affiliations by city.

Table 5
Most relevant journals and conferences.

Journal/Conference	TC	NP	Q
Journal of Medical Internet Research	1566	12	Q1
PLoS ONE	1117	12	Q1
Computers in Human Behavior	572	12	Q1
Journal of Biomedical Informatics	352	12	Q1
Procedia Computer Science	157	10	–

NP = number of publications, TC = times cited, Q = JCI Journal quartile.

4.4. Fake news in healthcare

In this section, the impact of fake news in the prediction in healthcare with Twitter analytics is evaluated. Applying algorithm 1 to the corpus of abstracts and titles for the publications in the scope of this study, the graph for fake news in healthcare is built in Fig. 8 and filtered by the combination of the ego-networks of the nodes "fake", "rumour", "trust", "misinformation", "disinformation", "veracity", "spam", "bot" and "privacy" with the goal of identifying the publications that study the diffusion of misinformation and how it affects healthcare.

After the network analysis applied to the resulting graph in Fig. 8, the categories identified and the most relevant publications in the scope are as follows.

- Disease outbreak, epidemiology, vaccination, and Covid-19: The fake news generated on Twitter spreads rapidly and, whether produced consciously or unconsciously, may negatively affect the individuals exposed to it or, even worse, massively impact society. This social problem, previously addressed in different domains like politics, has become more dangerous during the Covid-19 pandemic, as Twitter was by those who disseminated misleading information and conspiracy theories. During the pandemic outbreak, reliable information was scarce about how to protect against this disease, so fake news contributed to the confusion around the measures required to reduce the propagation of the coronavirus (in reality, the importance of wearing a mask, keeping a safe distance and receiving the vaccine). Exposure to this fake news and unreliable information put the health and safety of patients at risk, jeopardising an effective response to the crisis (Yousefinaghani et al., 2021). In this line, Madani et al. (2021) identified a total 37% of a real-time extraction from Twitter using various machine-learning algorithms was fake news. This illustrates the scale of the risk of misinformation and the need for critical thinking and reliable information sources, and identifies a positive correlation between negative and neutral sentiment and fake tweets. In the context of the Covid-19 pandemic, the propagation of fake news around vaccination awareness and its effectiveness has been disturbing. Hussain et al. (2021) applied artificial intelligence and sentiment analysis algorithms to confirm the usefulness of Twitter as a tool for real-time evaluation of vaccination awareness and discovering a positive correlation between negative sentiment and anti-vaccination or vaccine hesitancy and refusal.
- Fake news identification and veracity analysis: The need for reliable information to counter the increasing amounts of fake news and misinformation on social platforms has led to the development of "fact-checkers": content producers, organisations and individuals whose goal is to reduce the spread of misinformation. The Covid19 pandemic saw a burst of information, which the WHO[1] called an "*infodemic*", an "*epidemic*" of information or an excessive abundance of content that made it difficult to find useful information and reliable sources. On this topic, Burel et al. (2021) explored the impact of "fact-checkers" in reducing misinformation. They found that their effectiveness varies between different demographic groups, with different levels of susceptibility, and that their ability to reduce the spread of misinformation overall was limited.
- Techniques for misinformation detection: The existence of malicious content in social platforms means that mechanisms for identifying it need to be developed, with two aims. These are to minimise its impact and better understand how it is generated and spread. Wani et al. (2020) developed a stance detection system based on the combination of CNN (Convolutional Neural Network) with BERT (Bi-directional Encoder Representations for Transformers) for fact-checking in the context of the Covid-19 lockdown. Also, in the context of the need to analyse reliable data for efficient decision-making in healthcare, Sicilia et al. (2021) developed a novel machine-learning classifier for feature selection in Twitter datasets. Their model enabled them to detect rumours and misinformation in tweets with an accuracy of 96.8%, and therefore subsequently to focus their analysis on reliable information. They applied this technique to two different Twitter datasets: one about the Zika virus and one about vaccination awareness. In addition, the use of social network analysis for the identification of communities and the connections between individuals facilitates the discovery of those users who are creating misleading content, as in the case of attackers who maliciously compromised social accounts (Yen et al., 2021). This analysis also helps us understand the spread of fake news by exploring its propagation patterns (Kim et al., 2021). Finally, Ceron et al. (2021) demonstrated the effectiveness of combining sentiment

[1] https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200202-sitrep-13-ncov-v3.pdf.

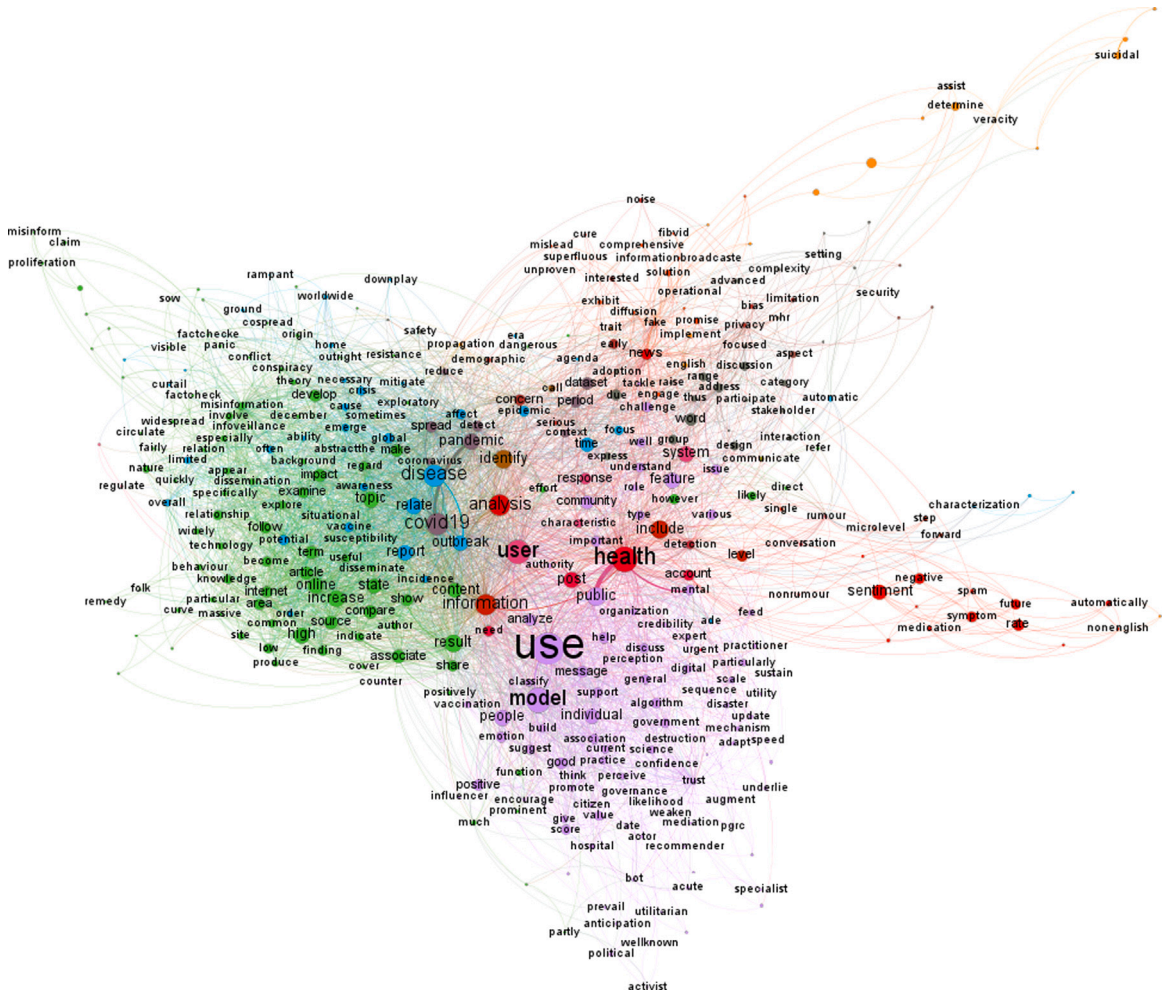

Fig. 8. Text-to-network analysis for fake news in healthcare.

analysis and topic modelling techniques to identify up to 30% of the content as misinformation and 21% of users as bots in 30 million tweets in a Covid-19 dataset.

5. Discussion

The social media have become an increasingly popular way for people to get information (Ozbay and Alatas, 2020). The healthcare industry has been widely evaluated in previous research. Therefore, given the many connections and social networks between physicians, patients and other interested parties, transparency and the open exchange of information take on special significance in social platforms. But fake news remains a serious problem, spreading false information increasingly widely (Gaurav et al., 2021). The proper use of Twitter can favour an environment of communication and collaboration for patients, physicians and researchers, and even improve the quality of care (Pershad et al., 2018). The ability to infer disease transmission networks and predict disease evolution are important challenges (Dhar et al., 2021), whose solution Twitter can assist via analysing transmission networks through user-shared content. Big data has grown worldwide and is especially important in an increasingly data-driven healthcare sector (Yan et al., 2019). The application of big data technology in healthcare is expected to continue to improve healthcare services, discover new insights and promote progress in healthcare (Chen and Liu, 2017). During 2020, there was a notable increase in the number of articles on the use of Twitter as a predictive system in the field of health. The present work has shown, through researching 492 articles from Scopus, Web of Science and ScienceDirect, that Twitter can be used as a predictive tool in the healthcare field and has pointed towards new lines of research exploring the future implications of the use of Twitter as a predictive tool. In addition, this in-depth analysis allows us to point to ways of detecting the integrity of information disseminated via social media. This has been especially important during Covid-19, and the vaccination period, because the proliferation of fake news in the healthcare field had devastating negative consequences (Herrera-Peco et al., 2021).

6. Conclusion

The results of this study, motivated by the gap in the healthcare literature research, contribute to the existing literature by providing an overview of the relevant insights extracted from the research using Twitter analytics as a predictive system in healthcare. This has been increasingly significant for both public health and mental health, helping the authorities identify disease outbreaks and cases of suicide or disorders of the individual. Our results also highlight the importance of awareness of, and preventive activities against, the rumours and fake news considered in the scientific literature in the field of health. To face this problem, fact-checking became a vital mechanism to fight false and malicious information. Our results highlight the importance of focusing analysis on how to identify relevant content and trustworthy sources and reduce the harm caused by misleading and malicious misinformation.

In general, the number of scientific articles has grown continuously since 2006, when Twitter emerged, and there was a remarkable

increase in interest in this topic upon the outbreak of the Covid-19 pandemic in 2020, which has reframed the concept of health. The users of social platforms such as Twitter have shared their impressions, uncertainties, fears and opinions. The data from this platform are unstructured, but by processing them and applying big data techniques which transform these analytics into insights, researchers can use Twitter predictively. Thus, Twitter analytics as a predictive system in health-related fields has gained more and more significance.

The impact of the scientific articles has been assessed through the SLR and SNA methodologies. The most representative countries with the highest scientific production in this topic are the United States, United Kingdom and Canada, partly because most of the available research is developed in English, the official language in these countries. The most influential authors are Signorini, A., Segre, A. M., & Polgreen, P. M and Eysenbach, G., who were pioneers in the use of Twitter analytics in predicting healthcare, and have continued their research on this area in the following years. The trending topics of greatest interest are mental health (e.g. suicide detection) and, in recent years, the Covid-19 pandemic.

The exploration of how to develop machine and deep learning technologies and big data analysis in biomedical applications and healthcare has been directly influenced by the contribution made by social platforms like Twitter for its use for prediction. The data generated in this platform makes it possible to analyse the digital footprints generated in the digital environment and to measure what previously could not be measured. Understanding the content generated by the users will continue to be essential in tackling certain mental health-related problems such as suicides, depression or abnormal moods, or reducing the adverse effects of drug misidentification for patient safety, and overcoming the noise generated by fake news. The development of artificial intelligence mechanisms for preventing the spread of fake news and misinformation is a crucial part of the automatic extraction of the knowledge contained in social platforms and the generated social networks. And understanding how fake news is generated and how to minimise it is essential for better decision-making by public health organisations and healthcare companies.

6.1. Theoretical implications

The present study sheds light on the main research areas and the practical contributions of the use of public Twitter analytics in the research field of healthcare. In addition, specific gaps in existing research literature have been identified.

The paper contributes to the evolution of the methodology of performing a systematic literature review, making it more exhaustive and thorough by leveraging machine learning and graph algorithms to conduct a massive evaluation of manuscripts.

6.2. Practical implications

The use of Twitter analytics as a predictive system in healthcare is an increasingly important tool supporting traditional models of user data collection and is therefore expected to contribute to changes in the healthcare sector.

In addition, Twitter is a publicly available data source that mirrors social perspectives across domains, and with the recent launch of the new academic API, historical public data is available. This facilitates the development of predictive studies that analyse the evolution of different topics and areas of knowledge using machine learning, Natural Language Processing and graph algorithms.

Given the amount of information shared by users every day on social platforms, it becomes essential for researchers and stakeholders to identify the emerging latent topics and social perspectives, distinguishing them from the generated misinformation and fake news, in order to extract relevant, high-quality insights. Therefore, professionals in healthcare, government policymakers and all related stakeholders should be encouraged to explore the potential of this tool and use it for the benefit of society and, particularly, patients.

6.3. Limitations

The main limitations of the study derive from its being a starting point for further scientific research. Firstly, the search terms conditioned the analysis, and the results may be different if new keywords and other academic databases are included. Second, given the growing scientific production generated in this field from 2021 onwards, new works can be anticipated, bringing new advances and updates of the concept that must be considered. Thirdly, the authors' criteria for identifying the publications in the scope of the study was determined by the manual screening process, which impacted the scope of the analysis.

Fourthly, because we cannot completely trust all information collected from Twitter, the results of analytics for predictive purposes need to be complimented by inputs from experienced healthcare professionals. Lastly, as a limitation generally in the overall field of these analytics, they reflect the reality of Twitter, not the entire society. Caution is therefore urged when framing generalisations on the conclusions about further research.

6.4. Future lines of research

Health is a current issue of major interest for institutions and governments; therefore, new studies using Twitter's capacity as a predictive system in healthcare should be considered. The growth of scientific production in this area will allow governments, institutions and companies to consider the data produced by Twitter users – real and fake – as a source of influential and updated social data across various areas of interest. Future lines of research will allow new correlations to be posed and in-depth analysis, including on the exploitation and functionalities of other social platforms such as Facebook, LinkedIn and Instagram. Such work will need to consider the various platforms' terms and conditions and copyright regulations, which have not been considered in this analysis, and could comparisons between different user profiles and segments will be useful. Further research is also needed to explore the potential of Twitter analytics in several areas, for example adverse event detection, AI applied to healthcare and measuring the societal effects of misinformation in health-related topics. More widely, because Twitter analytics can be used as a predictive tool in different areas of knowledge like society and economics, future research might merge economic and health-related analysis to predict the evolution of the healthcare market (digital transformation, the future of work in healthcare, etc.). Lastly, the need remains to ensure that the aggregated data from Twitter analytics is not biased by the fake news and rumours spread on the platform, so future work developing more effective fact-checkers will be important, especially in healthcare where the lives of patients are put at risk.

CRediT authorship contribution statement

Enrique Cano-Marin: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Writing – original draft, Validation, Supervision. **Marçal Mora-Cantallops:** Conceptualization, Term, Investigation, Writing – review & editing, Validation, Supervision. **Salvador Sanchez-Alonso:** Conceptualization, Term, Investigation, Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The authors have no conflicts of interest to declare. All co-authors agree with the contents of the manuscript and there is no financial interest to report. The submission is original work and is not under review at any other publication.

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